

Can We Predict Who Leaves Before They Go?

ConnectTel Communications Ltd

Dataset: 25,000 customers | 36 Features | 5 Models Evaluated

41.43%

Churn Rate

5.59 Cr / year

Revenue at risk

10,357 customers churned

Avg ARPU: ₹450/month

• The Problem We Are Solving

“ConnectTel Communications Ltd” - a pan-India telecom operator - is experiencing a significant and sustained increase in customer churn. This customer attrition is negatively affecting the company’s revenue, market position, and future expansion.

01

Reactive Only

Customers are engaged ONLY after they initiate port-out or disconnection requests. No proactive system exists.

02

Silent Churn

High-value customers leave without explicit complaints or warning signals, making detection extremely difficult.

03

Data Blind Spot

No systematic use of the 36 available behavioral, billing and service features to predict churn before it happens.

• Dataset Overview

25000

Customer Records

36

Columns

0

Missing Values

41.43%

Churn rate

Demographics & Plan

- Age, Gender, Region Circle.
- Connection Type (4G, 5G, Fiber).
- Plan Type (Prepaid / Postpaid)
- Contract Type, Tenure.

Usage & Billing

- Monthly & Total Charges.
- Avg Data (GB/Month)
- Voice Minutes, SMS Count.
- Overage Charges, ARPU

Service Experience

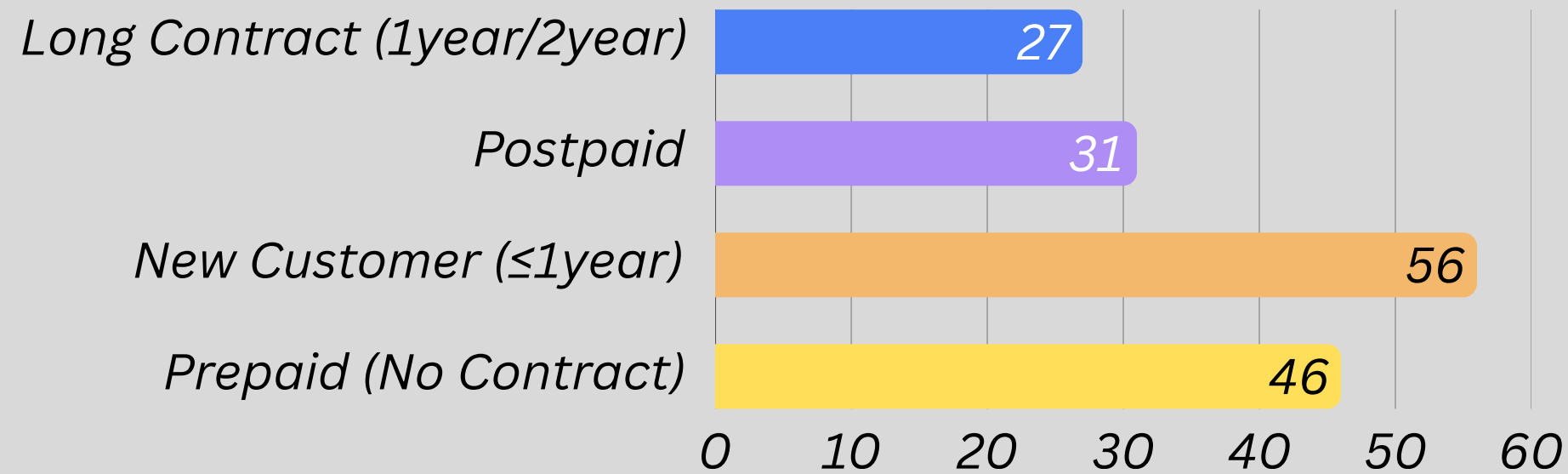
- Network Issues, Dropped Call Rate.
- Complaints Count (3m, 12m).
- Resolution Time (Days).
- Service and NPS Rating.

Payment & Competitor

- Auto Pay
- Payment Delays (Days)
- Competitor Offer received.

• What the Data Told Us – EDA Highlights

Churn Rate by Segment (%)



Prepaid Plan

70% customers

Highest Churn Segment

No Contract

17522 Customers

No commitment – Easy exit

Users

4G: 51.1% of customer base
5G: 30% of customer base

Better network – more stable

Postpaid Plan

29.9% customers

More loyal – Lower churn

Tenure

Avg tenure = 60 months

Fairly stable long - term customer base

Monthly Charges

Avg charge = ₹475/month

High variation (₹99/- – ₹1603/-)

New Customer Vulnerability

Customers within 12 months churn 2x faster.
Onboarding is critical

NPS Average = 20

Low, 'okay' customers are easy to lose. Strong churn predictor.

★ Service Rating = 3.8 / 5

Insufficient satisfaction across base.

20% Late Payers

Only 40% on autopay = weak behavioral commitment.

~8 logins/month

Low engagement. Self-care transactions ~2 per month

• Model Building Methodology

01 Data Loading & EDA

25,000 records, 36 columns. No missing values.
Churned: 41.43% customers (10357) & Retained: 58.57% (14643).

02 Feature Engineering

9 business driven features created, arpu & tenure months dropped from model. arpu due to multicollinearity and tenure converted into derived feature.

03 Train-Test Split

80/20 stratified split.
Train: 20,000 | Test: 5000
stratify=y ensures data balance preserved.

04 Encoding & Scaling

OneHotEncoder for 7 categorical features.
Robust scaler for numerics (outlier-resistant).

05 Model Training (5 Models)

GridSearchCV + 5-Fold Stratified cross-validation
Optimization metric: Recall (churn class).

06 Threshold Tuning

Precision-Recall curve used
Best Threshold 0.4154 → not recommended for deployment (too many false positives).

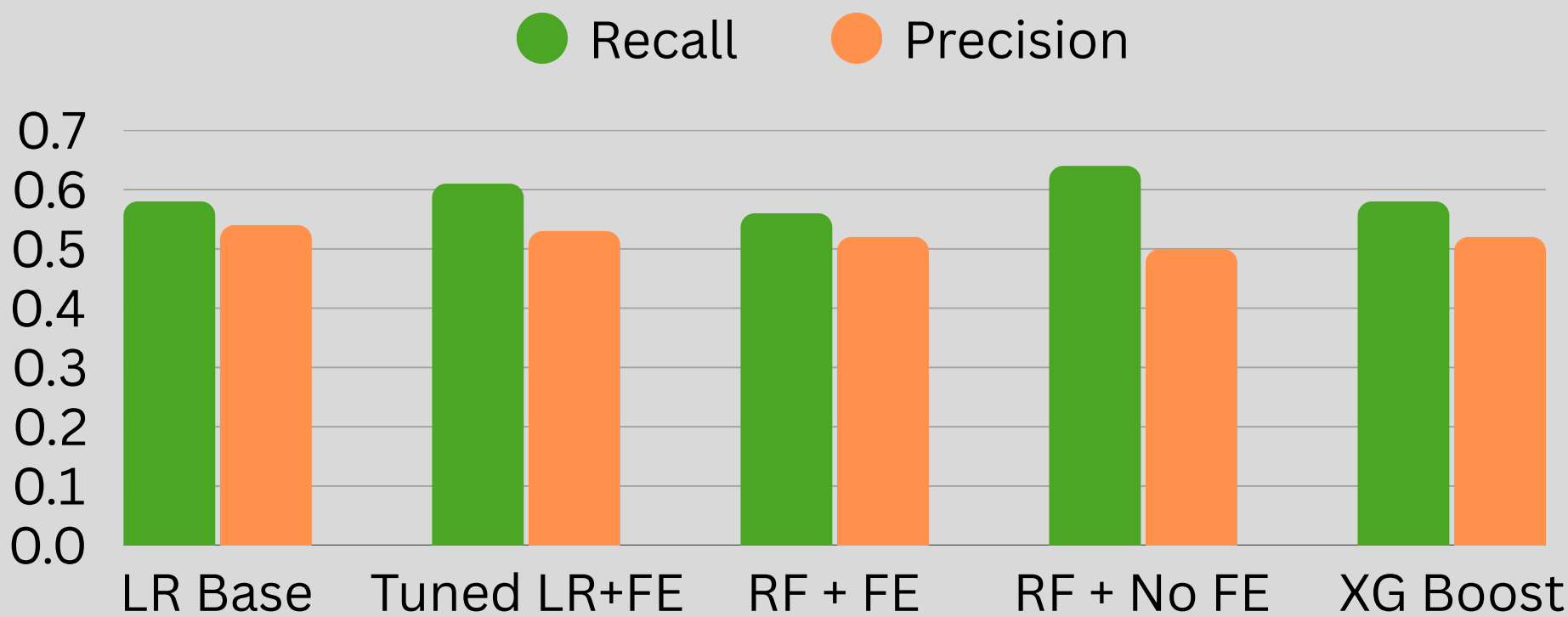
• Feature Engineering - 9 Business driven Features

9 new features were engineered to capture hidden behavioral patterns that raw data cannot express directly.

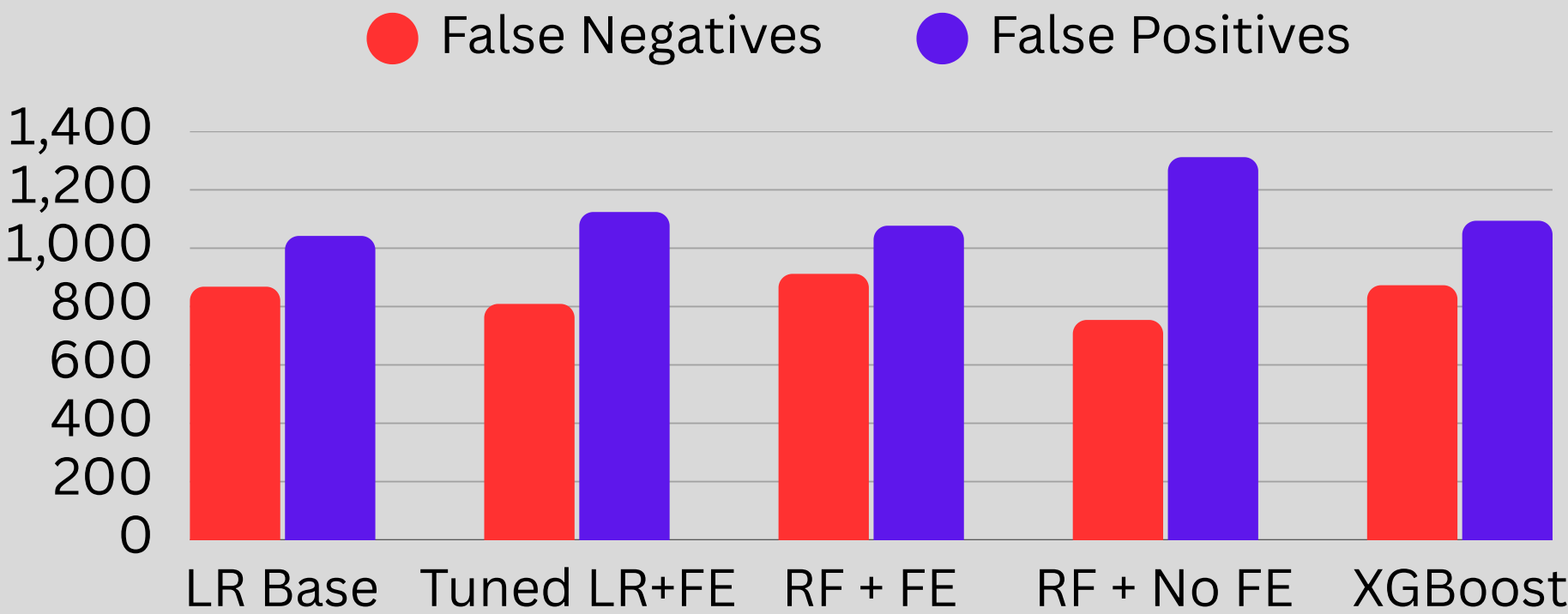
Feature Name	Business Rationale	Insight
high_data_consumption_flag	Top 25% data users. Heavy users are revenue critical.	Premium Retention Priority.
high_voice_usage_flag	Top 25% voice users. Identifies heavy engagement segment.	Cross-sell opportunity.
low_usage_flag	Bottom 25% in BOTH data and voice usage. Low engagement.	Engagement campaign.
lifecycle_stage	New(<= 12m) / Early(13-24m) / Mature(25-60m) / Loyal(>60m)	Lifecycle specific strategies
poor_complaint_handling	Recent complaints + resolution_time > 2 days. Service failure + inefficiency.	Trigger urgent escalation.
new_customer_high_risk	tenure <= 12 months & plan type = prepaid.	New customer with prepaid plan indicates churn risk.
nps_category	Detractor (nps_score < 0) / Passive (0 - 50) / Promoter(score > 50)	Convert subjective feedback to structured churn predictor.
low_service_rating	Bottom 25% service rating.	Dissatisfaction and high-churn prob.
plan_fit_score	Inverse of the ratio of overage charges to monthly charges.	Low score - Customers frequently pays extra - poor-plan fit.

Model Comparison - Performance Metrics

Recall Vs Precision Comparison



FN vs FP Count

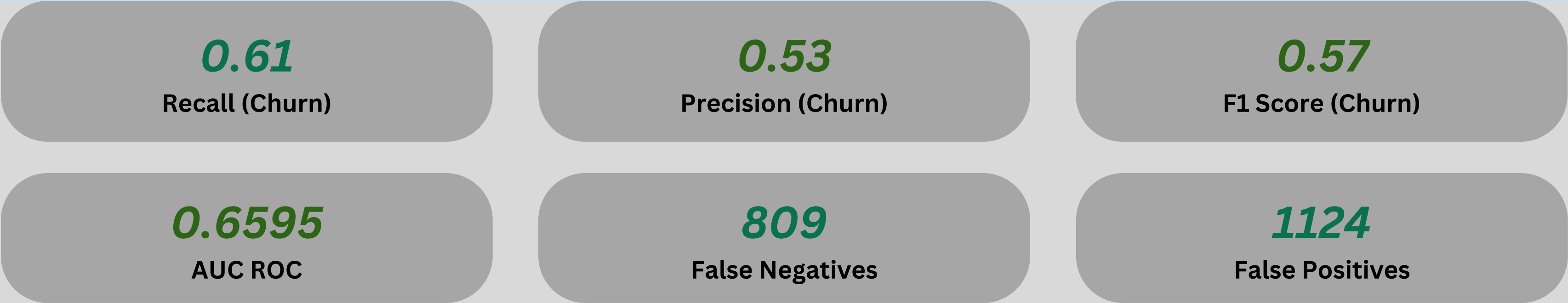


Model	Precision	Recall	F1	ROC-AUC	FN	FP
Baseline LR (with FE)	0.54	0.58	0.56	0.661	868	1042
Tuned LR (with FE) (Selected)	0.53	0.61	0.57	0.659	809	1124
Random Forest (with FE)	0.52	0.56	0.54	0.637	912	1077
Random Forest (without FE)	0.50	0.64	0.56	0.639	754	1312
XG Boost (without FE)	0.52	0.58	0.55	0.655	873	1094

• Final Model – Tuned Logistic Regression (with FE)

 SELECTED MODEL: Tuned Logistic Regression with Feature Engineering

Best Params: C=0.01 | Penalty=l1 | solver=liblinear | class_weight=balanced | Best cv Recall=0.6288

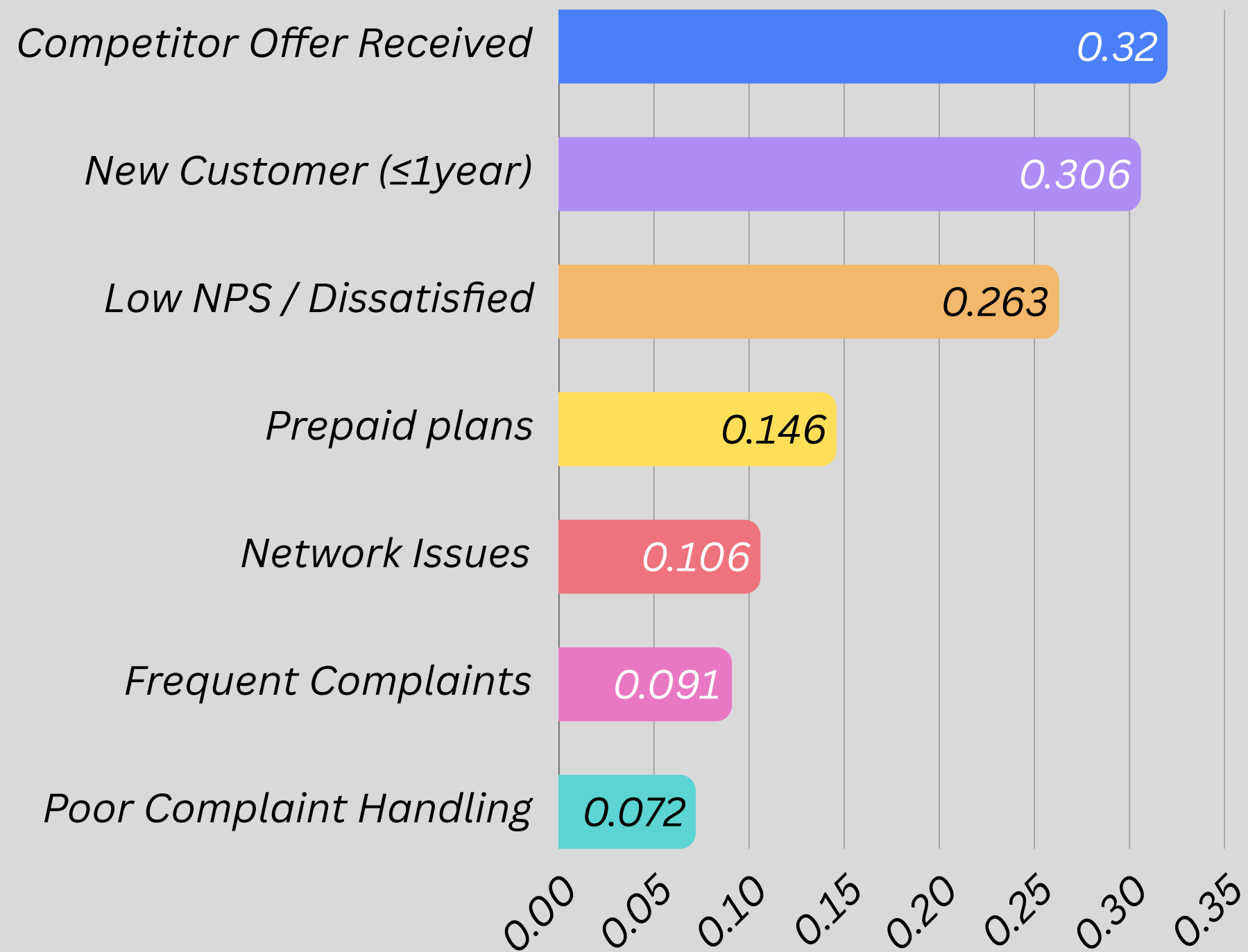


Segment Wise Performance

Segment	Model	Precision	Recall	Model	Precision	Recall
Low Value	LR (with FE)	0.53	0.74	RF (no FE)	0.49	0.83
Medium Value	LR (with FE)	0.54	0.52	RF (no FE)	0.53	0.48
High Value	LR (with FE)	0.46	0.17	RF (no FE)	1.0	0.02

• Churn Drivers & Threshold Tuning Analysis

• Top Churn Drivers (LR Coefficient)



• Top Retention Signals

High Service Rating & NPS Score

Postpaid Plan Subscribers

Long Tenure (> 24 Months) / High Total Charges

Threshold Tuning result: Best threshold = 0.4154 → Recall jumps to 0.86 BUT Precision drops to 0.47, causing excessive false positives, NOT deployed. Default threshold (0.5) used with recall = 0.61.

• Strategic Recommendations

A

Deploy as Decision-Support in CRM

Integrate model for monthly/weekly scoring AUC=0.66 supports decision making, not full automation. Business judgement remains essential

B

Risk Based Customer Tiering

Critical(>0.60): Aggressive personal outreach, high-value discounts. High(0.50-0.60): Strong offers + complaint resolution. Medium (0.45-0.50): Emails, loyalty rewards. Low (<0.45): No action.

C

Top-N Targeting for High-Value Segment

Model recall for high-value is only 17%. Use top-30% probability ranking within high-value segment. 43.18% churn density vs 31.35% overall → higher ROI per campaign rupee.

D

Address Root Cause Churn Drivers

Counter competitor offers proactively. Fix onboarding for new customers ($\leq 12m$). Reduce complaint resolution time (<2days). Upgrade network quality.

• Business Impact Quantification

10,357

Actual Churners

₹ 450/month

Average ARPU (churned)

₹ 5.59 Cr

Annual Revenue at Risk

6,317

Churners Identified by Model (61% Recall)

2,210

Retained (35% Conservative Rate)

₹ 1.19 Cr

Annual Revenue Saved

Revenue Risk Mitigated: 21.34% | ₹1.19 Cr saved from ₹5.59Cr at risk | Retention rate: 35% (conservative rate assumed)

Top N Targeting Strategy – For High-Value Segment (Test Set Results)

Target Group	Customers Targeted	Churn Captured	Churn Rate (%)
Top 30% (Recommended)	176	76	43.18
Top 50%	293	111	37.18
All Customers	587	184	31.35

• Conclusion & Key Takeaways



Churn is Predictable

Tuned LR (FE) achieves Recall=0.61 identifying 6,317 out of 10,357 churners for targeted intervention.



Root Cause are Clear

Competitor offers, weak onboarding (<12m), and low NPS are the top 3 churn triggers – all addressable.



Model Selected Correctly

Tuned LR beats RF on segment precision, false positives control, and business explainability.



Data – Constrained

AUC=0.66 reflects dataset limits, not model failure, Even XGBoost shows similar performance.



Clear Path Forward

Richer data signals + quarterly refresh = significant future improvement.



Financial Impact Proven

₹1.19 Cr saved annually from ₹5.59 Cr at risk = 21.34% revenue risk mitigated.

Thank You